

CO4-AT1-Industry Problem Solving

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1. Question

Q1 INDUSTRY TASK 1: Employee Performance & Payroll Dashboard

25 marks

Problem Statement: A software company wants to develop a simple Employee Performance and Payroll Management System using Java AWT. The application should allow the HR team to enter employee details, process salary information, and generate a structured payroll report.

Functional Requirements

- Develop an AWT-based graphical user interface.
- Accept Employee ID, Employee Name, Department, Designation, Basic Salary, Working Days, and Performance Rating.
- Use suitable AWT components such as Label, TextField, Choice, Checkbox/CheckboxGroup, Button, TextArea, and List.
- Calculate Gross Salary, Performance Bonus, and Net Salary using suitable business rules defined by the student.
- Display the complete employee payroll report in a List or TextArea when Generate Report is clicked.
- Use GridBagLayout or BorderLayout for arranging the interface.
- Implement button events using ActionListener.
- Provide Submit/Generate Report and Reset functionality.
- Validate important inputs such as empty fields and numeric salary values.

Industry Context: The application simulates an HR payroll module used to capture employee information, perform basic payroll processing, and present a report for HR verification.

Expected Deliverables

- Java source code.
- Working GUI application with event handling.
- Sample input and output.
- Screenshot of the final application.
- Brief explanation of components, layout manager, and event handling used.

Answer:

The screenshot displays the Eclipse IDE environment. On the left, the Package Explorer shows the project structure with files like AppletLifeCycle.java, BorderLayoutDemo.java, CalculatorApplet.java, CardLayoutDemo.java, EmployeePayrollAWT.java, GridCalculator.java, GraphicApplet.java, StudentProfile.java, and StudentRegistration.java. The main editor shows the source code for EmployeePayrollAWT.java, which is a Java AWT application. The code includes imports for java.awt.*, java.awt.event.*, and java.util.*. It defines a public class EmployeePayrollAWT that extends Frame and implements ActionListener. The class contains fields for employee details (Employee ID, Name, Department, Designation, Basic Salary, Working Days, Performance Rating) and a TextArea for the report. The main method sets the title to 'Employee Performance & Payroll Dashboard', sets the size to 650x650, and sets the layout to BorderLayout. It creates a new Label for the title and a new Panel for the form. The form is a GridBagLayout with GridBagConstraints. It includes text fields for Employee ID, Name, Department, Designation, Basic Salary, and Working Days. It also includes a Choice box for Department and a Choice box for Designation. The form is populated with sample data: Employee ID: 101, Employee Name: K.Dhilip, Department: HR, Designation: Team Leader, Basic Salary: 75000, Working Days: 28, Performance Rating: Excellent. The form is displayed in a window titled 'EMPLOYEE PERFORMANCE & PAYROLL DASHBOARD'. The window contains a 'PAYROLL REPORT' section on the right, which displays the calculated payroll details: Employee ID: 101, Employee Name: K.Dhilip, Department: HR, Designation: Team Leader, Working Days: 28 days, Performance Rating: Excellent, Basic Salary: ₹ 75000.00, Allowance (20%): ₹ 15000.00, Gross Salary: ₹ 90000.00, Performance Bonus: ₹ 11250.00, Deduction (5%): ₹ 4500.00, NET SALARY: ₹ 96750.00. The window also has 'Generate Report' and 'Reset' buttons.